LESSON PLAN

LSF, STABILITY, A/C PERFORMANCE (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	10
Total No. of Lesson Plans	33	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Video demonstrating fluid flow characteristics
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Video demonstrating fluid flow characteristics. Video on theory of flight Lift (b) Drag (c) control.
5. Specific Objectives	6.1.1.1 Trainee will be able to explain the properties of atmosphere, flow over bodies, boundary layer.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review video segments; Instructor supervises note-taking on flow separation and boundary layer concepts.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the three primary forces acting on an aircraft in flight?
(b) Define the boundary layer in the context of fluid flow over a body.
(c) List two properties of the atmosphere relevant to aerodynamic performance.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Properties of Atmosphere • Flow Over Bodies • Boundary Layer Formation • Flow Separation • Viscosity Effects	L & Ds	PPT, IB	15 min
(b)	• Lift Generation Principles • Drag Components Analysis • Control Surface Mechanics • Pressure Distribution • Angle of Attack Effects	L & Ds	PPT, PDF	20 min
(c)	• Interrelation of Forces • Stability Characteristics • Performance Implications • Flow Visualization • Real-world Applications	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the boundary layer affects drag on an aircraft body.
(b) Describe the relationship between angle of attack and lift coefficient.
(c) Identify the primary control surfaces used for pitch, roll, and yaw.

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of lift and drag discussed in this lesson will be expanded upon in LP 11 regarding factors affecting the coefficient of lift.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Factors affecting CL, Theories of Lift production, force and pitching moment, coefficient of lift, variation of co-efficient of lift with aspect ratio and sweep back and CL.	Refer to the master syllabus reference list	LP 11

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min